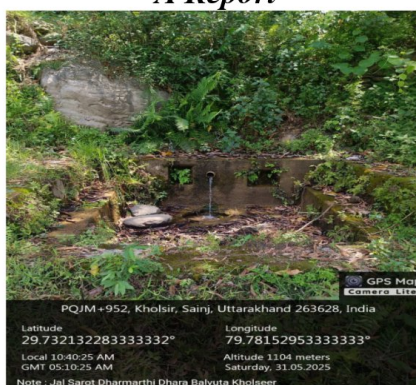


REJUVENATION STRATEGY OF THE BHAILYUTA SPRING & PANYALI GADGERA VILLAGE BHAILYUTA DISTRICT BAGESHWAR IN JAIGAN WATERSHED

A Report



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Sponsored by:

AMAN

Eastern Pokherkhali Almora

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REJUVENATION STRATEGY OF THE BHALYUTA SPRING IN PANYALI GADGERA WATERSHED,VILLAGE BHALYUTA IN JAIGAN WATERSHED,DISTRICT BAGESHWAR

1.0 INTRODUCTION

Skilled human resource and sufficient funds are needed for reversing steady transformation naulas, springs and streams for their perennial nature to ephemeral nature. There is acute scarcity of both. In view of this acute scarcity of skilled human resource and sufficient funds, the work of rejuvenation of the Panyali Watershed water resources has to be carried out in different phases. In the first phase – as per the need of the local inhabitants of village Balyuta, a GIS based strategy has been developed by delineating surface recharge zone to rejuvenate the dying Balyuta springs and Panyali stream.

1.1 About the Bhalyuta Spring

The Baluyata spring ($29^{\circ}44'4.41''\text{N}$ Lat and $79^{\circ}46'54.34''\text{E}$ Long) lies at the height of 1170m from the msl in the northern part of the lesser Himalaya experimental recharge zone area.

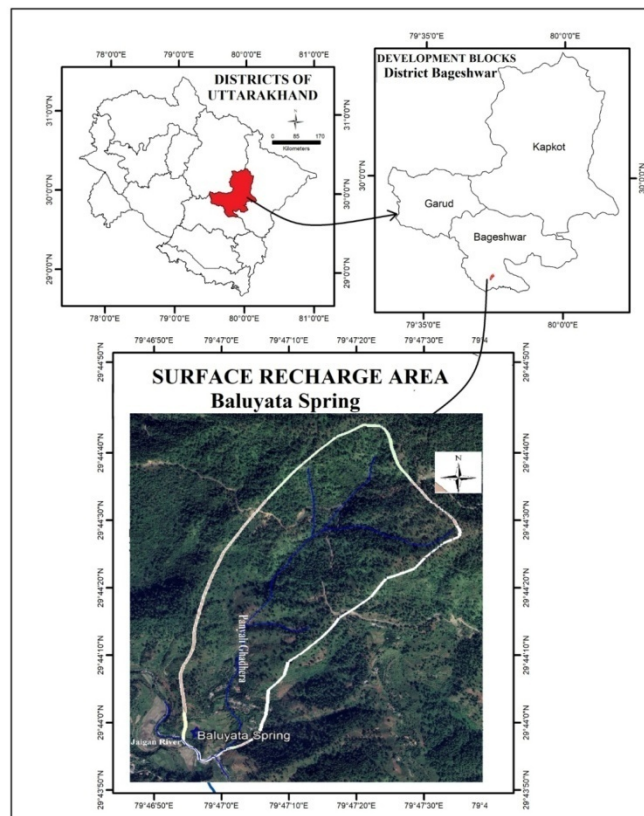


Fig.1: Location map of the Baluyata Spring and Panyali Gadhera and their surface recharge area.



Fig.2: The Panyali Watershed in village Balyuta, District Bageshwar in Jaigan Valley.

The study area, viz., the Panyali Watershed lies in between $29^{\circ} 43'52.158''$ N to $29^{\circ} 44'47.547''$ N Lat and $79^{\circ}46'51.248''$ E to $79^{\circ}47'36.248''$ E Long in the Lesser Himalayan rain fed river regime in district Bageshwar (Fig.2) which encompasses an area of 0.95km^2 . The altitude in this watershed varies in between 1100m to 1800m. Climatically, the watershed falls in cool temperate and cold temperate climatic regions. Geologically, the watershed is composed of metamorphosed and sedimentary rocks of the Damta group (Valdiya,1985)

1.2 Land use Pattern

The land use pattern of the study area is presented in figure 3. Out of the total area, a large part, i.e., about 3.20% (0.03km^2) area is under agricultural land which is now mostly in abandoned form due to migration of people from the village. About 53.11% (0.50km^2) area falls under barren land. About 0.41km^2 area, which constitutes about 43.01% of the total area is under forest/plants. The settlements in the recharge area cover about 0.35% (0.001km^2) area while about 0.84% (0.01km^2) area is covered by road.

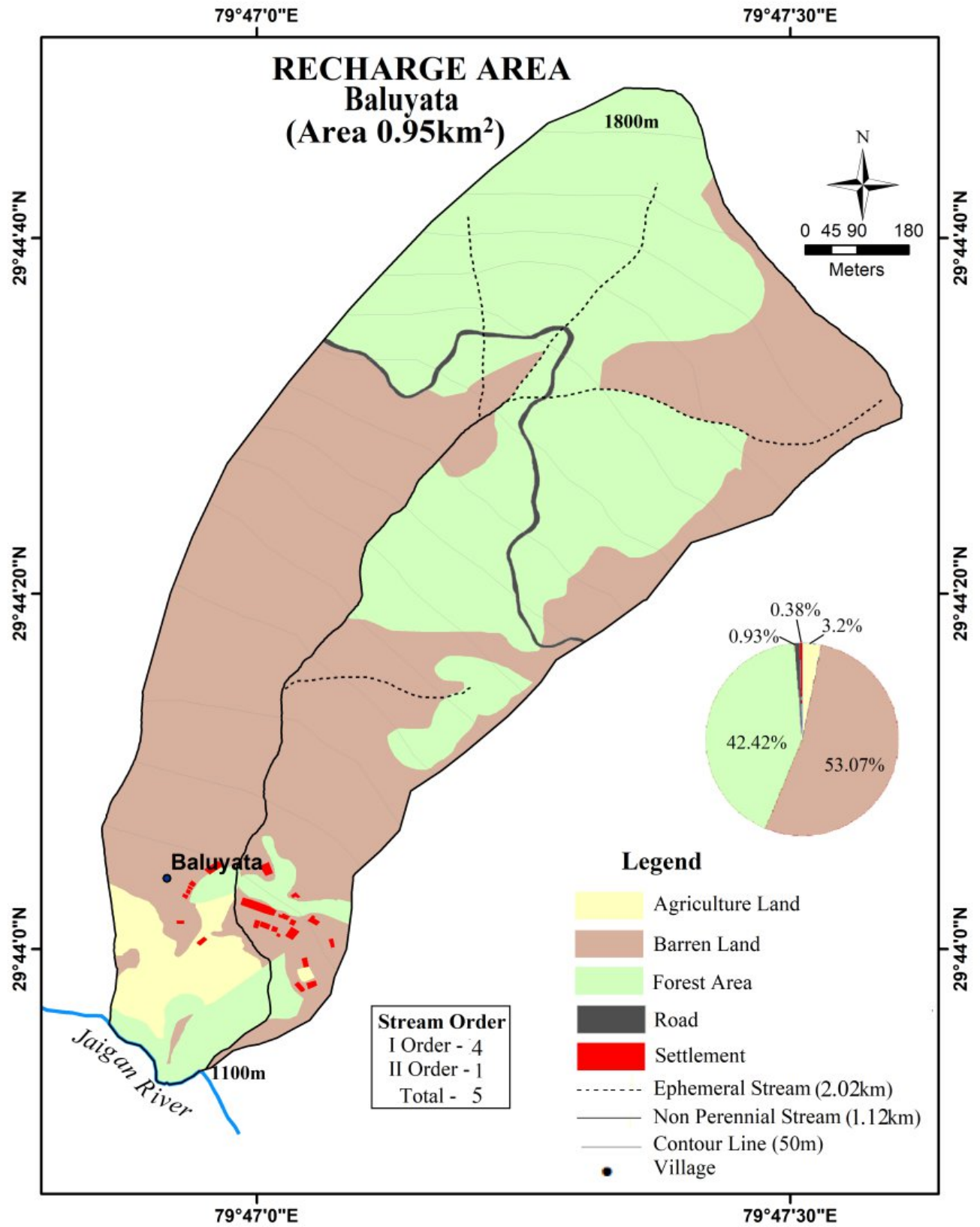


Fig.3: Land use pattern in the study area

2.0 REJUVENATION TREATMENT MEASURES

The treatment measures to be taken for rejuvenation of the spring, naulas and streams (*gadgeras*) of the experimental area are divisible into two different types These are mechanical treatment measures and biological treatment measures. A brief account of objectives of these rejuvenation measures is presented in the following paragraphs..

2.1 Mechanical Treatment

The biological treatment measures take longer time to improve the condition of land for rain water infiltration, therefore, for speedy groundwater augmentation, mechanical treatment measures are needed in the recharge area. The objectives and types of mechanical treatments to be taken are described in the following paragraphs.

2.1.1 Objectives

Particular Objectives

1. Groundwater augmentation by increasing the rainwater percolation capacity of bed rocks to recharge groundwater storage.
2. To increase soil moisture storage in the top-soil, sub-soil and parent material horizons of land.
- 3 To control surface runoff or the problem of flooding;

General Objectives

1. To control accelerated soil erosion;
- 2 To reduce dissolved, suspended, and bed load concentration of streams/ rivers water.
- 3 To bring back *hariyali* on the nature through water conservation;
- 4 To improve the quality of economic environment of the local environment.

2.1.2 Treatment Measures

To complete above objectives, the following six structures are recommended for mechanical treatment to fetch the rainwater for groundwater augmentation to recharge springs, naulas and streams. These are: infiltration holes, infiltration trenches, bio-plugs, rill/gully plug, contour bunds, terrace ponds and check dams. A brief accounts of these structures is presented below.

2.1.2.1 Infiltration Holes (Plate-I)

These are small holes (size 10-15 cm deep, 5cm dia and density 5 per/m²) as shown in *Plate-I*, to be constructed on the degraded forest and barren lands to catch and infiltrate the rain water of the high intensity rain storms into the soil for groundwater augmentation.



Plate-I: Example of infiltration holes (size 10-15 cm deep, 5cm dia and density 5 per/m²) constructed to fetch high intensity rain drops in barren as well as in degraded forest to facilitate infiltration of rainwater for groundwater augmentation.

2.1.2.2 Infiltration Trenches (Plate-II)

These are small cuts (size 1.0 x 0.5 x 0.3 m) as shown in *Plate-II*, on the hill slopes made at a spacing of 10m at an average slope condition of 20°. The distance of spacing may be decreased, in case the hill slope is low. These should be constructed in stagger pattern. The purpose of these infiltration trenches is to catch the excess hill slope runoff and to facilitate it for groundwater recharge. These infiltration trenches should not be constructed on discharge zones and flat lands.



Plate-II: An example of infiltration trench (size 1.0 x 0.5 x 0.3 m) to trap the overland Sheet flow from the hill slope for infiltration and also to trap the sheetwash erosion sediment.

2.1.2.3 Bio- Plugs (Plate-III)

These are small bio-plugs to be constructed by vegetation and dead wood logs or pine needles and soil across an ephemeral rill/gully. The purpose of these bio-plugs is to break the speed of runoff in the ephemeral rill/gully and to store rain water to facilitate it for groundwater recharge (Plate-III). The size of these bio-plugs depends upon the local terrain condition.



Plate-III: An example of bio-Check dam constructed using pine needle across a first order ephemeral stream to break the velocity of runoff, to store water for infiltration in to the aquifer and to check the erosion.

2.1.2.4 Rill Plugs/Gully Plugs (Plate-IV)

Due to high intensity rain events, the hill slopes are being eroded steadily through the runoff along ephemeral channels by which deep gullies and rills are being developed. To control this process of gullying/rilling, there is need to construct plugs of stones across these rills/gullies to decrease the velocity of streams, to store rainwater for infiltration and to trap the sediment. The size of such plugs (Plate-IV) depends upon the local terrain condition.



Plate-IV: An example of gully plug across a first order ephemeral stream to check runoff velocity, to store the rainwater and to control process of erosion.

2.1.2.5 Check Dams (Plate-V)

These are series of small check dams, constructed using cement and iron across second/third order streams having a height of 1.0 m to 1.5m (depending upon the local terrain condition of the site). The purpose these check dams constructed across stream at an interval of 30m to 50m (depending upon slope condition) is to check the flow of intermittent/seasonal streams to facilitate it for groundwater recharge, and to mitigate the process of erosion.



Plate-V: An example of series of check dams across an intermittent/seasonal stream to collect water for infiltration and irrigation purposes.

2.1.2.6 Contour Bunds (Plate-VI)

The contour bunds are earthen embankments along contours to capture rainwater in agricultural land (Plate-VI). These help in storing the rainwater for infiltration to groundwater augmentation, reduce erosion and also help in irrigating agricultural crops. These contour bunds should develop in association with the local residents.



Plate-VI: An example of series of contour bunds constructed on the edges of agricultural fields to trap rainwater for irrigation as well for infiltration to recharge groundwater storage.

2.1.2.7 Terrace Ponds (Plate-VII)

The terrace ponds are small, artificial water tanks constructed on terraced fields in gentle hill slope areas. These are small shallow tanks with earthen embankments. Their size varies, but typically 10-20 m² in area and 1-2 m deep. The Dig pond area, using The excavated soil of the dig pond are should be used for construction of embankments. They're designed to collect and store rainwater through nearby channels or pipes bring runoff from surrounding catchment areas runoff, providing water for recharge in to the aquifer, irrigation and other uses. They also help to control erosion in the hills.



Plate-VII: An example of terrace pond for rainwater harvesting to groundwater recharge and irrigation of different crops.

2.1.2.8 Building Rainwater Harvesting for Domestic Use (Plate-VIII)

Rainwater harvesting systems collect and store rainwater from rooftops for domestic use, reducing dependence on external water sources. This system has 4 components:

- i) Roof top collection system: Gutters and downspouts collect and direct rainwater
- ii) Filtration unit: Removes debris and contaminants from the collected water
- iii) Storage tank: Stores the filtered water for later use
- iv) Distribution system: Supplies harvested water for domestic purposes (e.g., flushing toilets, washing clothes, cleaning, irrigation)

This rainwater harvesting system has the following benefits:

- i) Water conservation: Reduces water bills and dependence on external sources
- ii) Reduced storm water runoff: Decreases burden on drainage systems
- iii) Non-potable use: Suitable for uses that don't require treated water
- iv) Eco-friendly: Promotes sustainable water management practices.

This rain water system may be used for domestic uses such as: i) flushing toilets, ii) washing clothes, iii) cleaning, iv) irrigation, and v) washing utensils /vehicles etc.

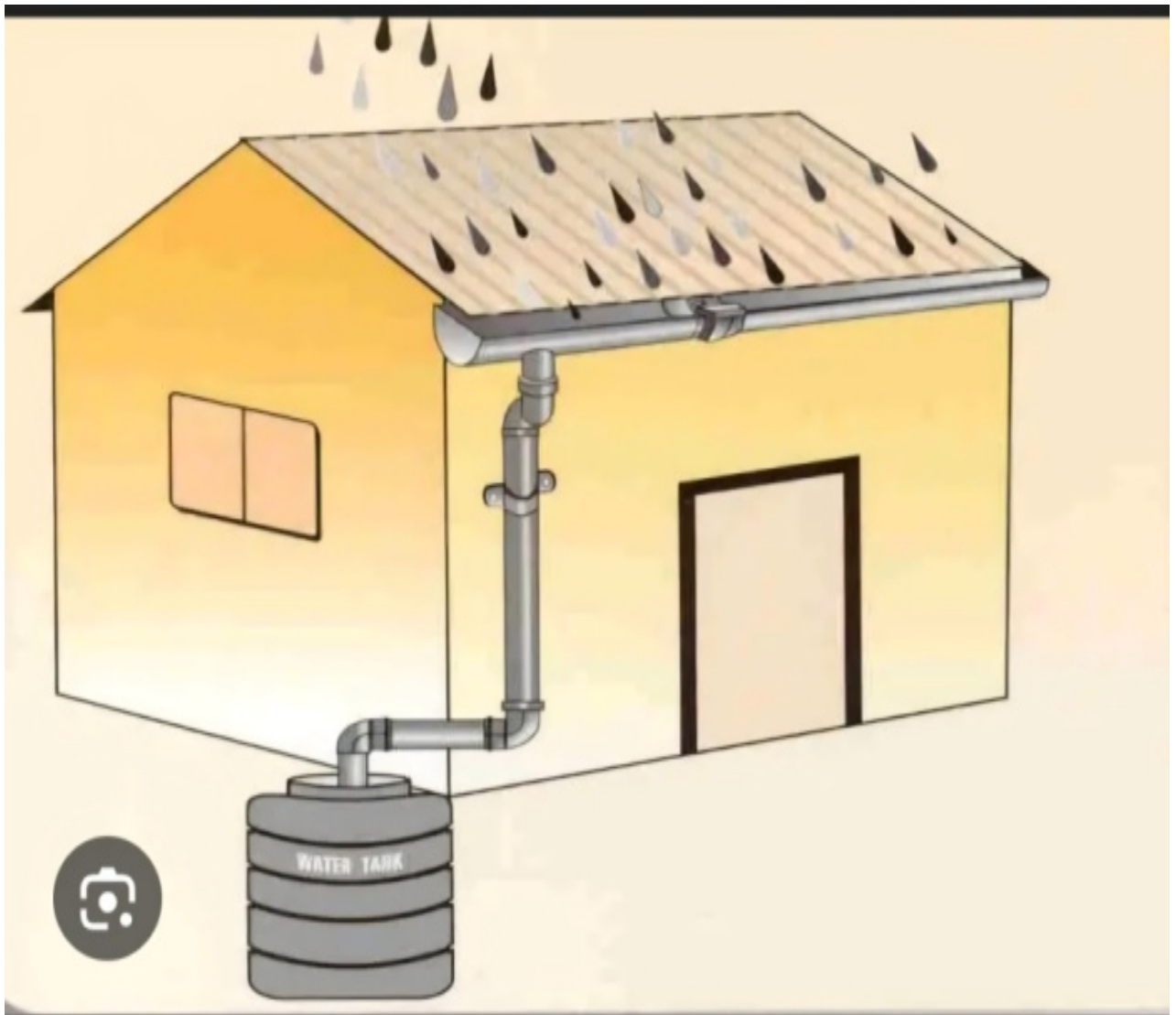


Plate-VIII: Rooftop rainwater harvesting structure for kitchen gardening and utensil cleaning etc.

2.1.2.9 Building Rainwater Harvesting for Groundwater Augmentation (Plate-IX)

Building roof rainwater harvesting structures in hills can help in rejuvenating dying springs and streams by augmenting groundwater resources. The system captures rainwater from rooftops, stores it, and allows it to percolate into the ground, recharging aquifers (Plate-IX). This system has the following 4 components (Plate-IX).

- i) Rooftop collection system: Gutters and downspouts collect and direct rainwater
- ii) Filtration unit: Removes debris and contaminants from the collected water
- iii) Storage tank: Stores the filtered water for later use
- iv) Recharge pit: Allows excess water to percolate into the ground, recharging aquifers.

The benefits of this system are: i) rejuvenates dying springs and streams, ii) augments groundwater resources, iii) reduces soil erosion, iv) conserves water, and mitigates water scarcity in hilly areas.

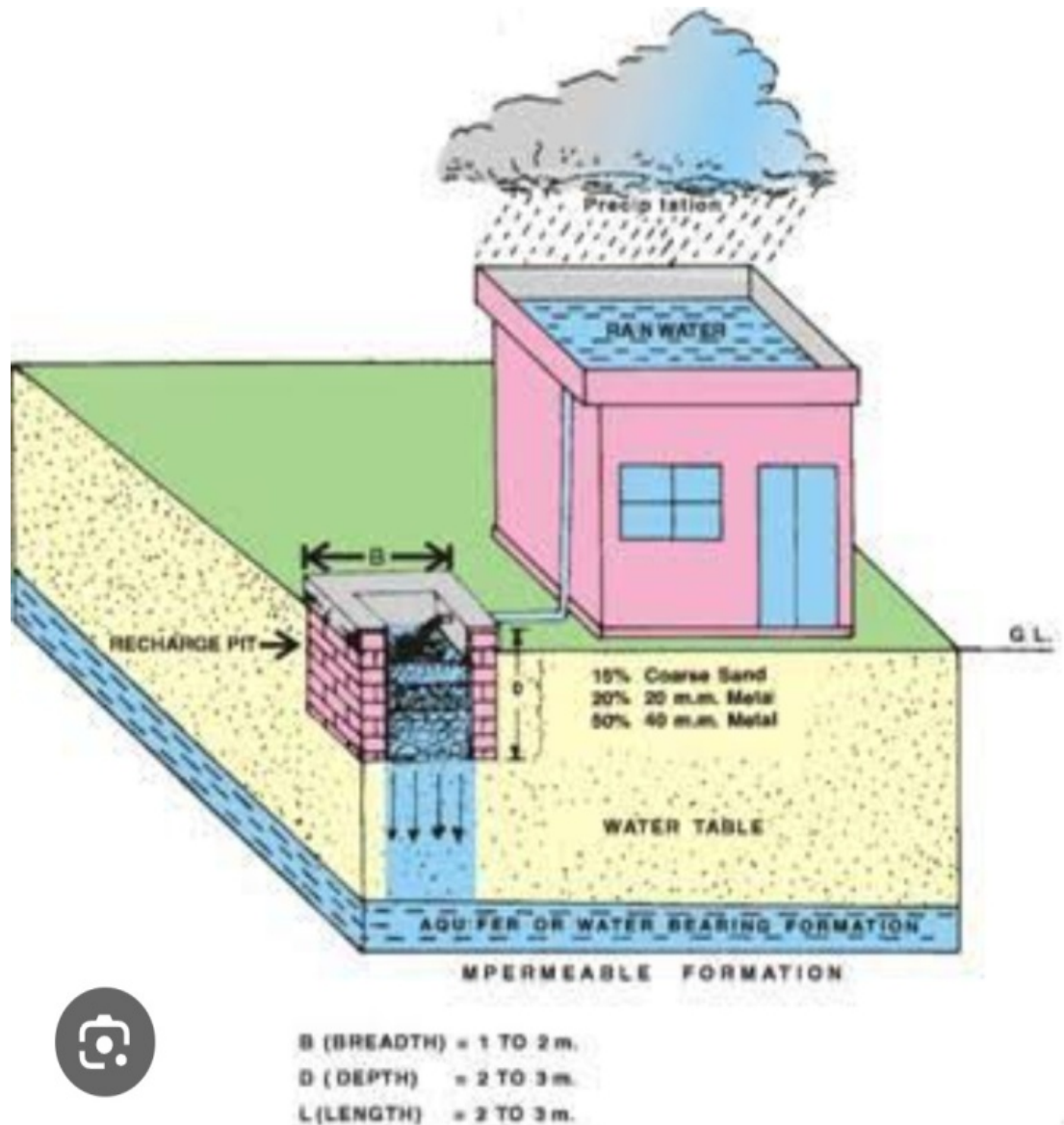


Plate-IX: Rooftop rainwater harvesting structure for groundwater

2.1.2.10 Rainwater Harvesting for Groundwater Augmentation and Domestic Use (Plate-X)

Building rainwater harvesting structures in hills can serve multiple purposes also, including domestic use, groundwater augmentation, and rejuvenation of dying springs and rivulets. The system captures rainwater from rooftops, stores it, and utilizes it for various needs. This system has five components (Plate-X):

- i) Rooftop collection system: Gutters and downspouts collect and direct rainwater
- ii) Filtration unit: Removes debris and contaminants from the collected water
- iii) Storage tank: Stores the filtered water for later use
- iv) Recharge pit: Allows excess water to percolate into the ground, recharging aquifers.
- v) Distribution system: Supplies harvested water for domestic purposes (e.g., irrigation, toilet flushing, cleaning).

There are different benefits of this system as mentioned below.

- i) Domestic use: Provides water for non-potable purposes, reducing dependence on external sources
- ii) Groundwater augmentation: Recharges aquifers, rejuvenating dying springs and rivulets.
- iii) Soil conservation: Reduces soil erosion and landslides
- iv) Water scarcity mitigation: Helps alleviate water scarcity in hilly areas
- v) Eco-friendly: Promotes sustainable water management practices,
- vi) Rejuvenation of Dying Springs and Rivulets: Recharged groundwater feeds into springs and rivulets, restoring their flow and
- vii) Helps maintain aquatic ecosystems and supports biodiversity

Thus, by implementing rainwater harvesting structures in hills, communities can ensure a sustainable water supply, conserve natural resources, and promote eco-friendly practices.

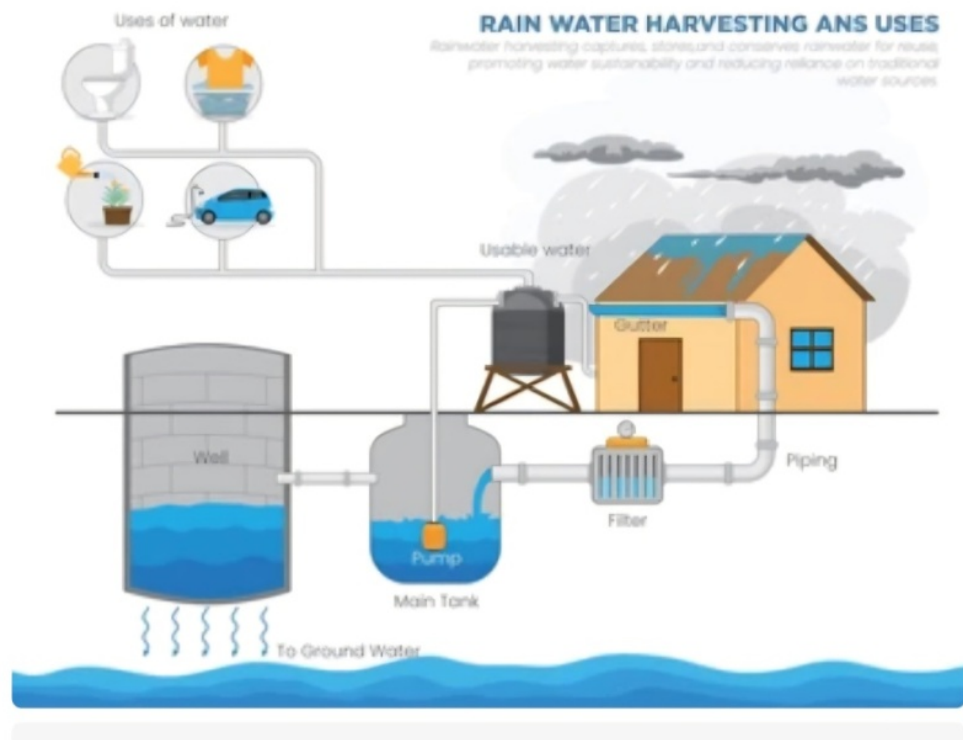


Plate-X: Rooftop rainwater harvesting structure for groundwater augmentation for rejuvenation of springs and for different domestic purposes

2.1.3 Biological Treatment

Biological treatment measures include –afforestation, plantation of trees, shrubs and grasses, and development of nurseries and horticulture. The past experiences reveal that, no biological treatment work should be done for rejuvenation of dead/dying water resources without ensuring the complete protection from fire as well as from damages through animals and human beings. A brief account of objectives and treatment measures of biological treatment is presented below.

2.1.3.1 Objectives

Particular Objectives

1. To increase the rainwater infiltration capacity of land for soil moisture storage.
2. To increase groundwater storage by deep percolation of rainwater into rocks on long term basis.
- 3 To decrease surface runoff.

General Objectives

1. To control accelerated erosion.
2. To mitigate the problem of flooding on long term basis
3. To control the process of masswasting, i.e., land creeping, slumping, land sliding etc,
4. To regenerate the degraded natural environment;
5. To improve the quality of economic environment of the local inhabitants..

2.1.3.2 Treatment Measures

In order to meet above objectives, the biological treatment measures such as afforestation, plantation, and development of horticulture and nurseries should be done. A brief accounts of all these biological treatment measures is presented in the proceeding paragraphs.

2.1.3.2.1 Afforestation

Afforestation means establishing a forest in an area where there was no previous tree cover. This involves planting of native species, allowing natural regeneration, promote biodiversity. This may be done on degraded land, like barren areas or abandoned agricultural lands. It is suggested that in such available areas afforestation of water harvesting (broad leaved) species, local shrubs and grasses such as Napier and vertebrae should be done by initiating massive social forestry programmes in the recharge areas with close association with local community and the Forest Department and Watershed Management Department.

2.1.3.2.2 Plantation

This is a process of establishing a forest plantation in an area often with a single species (monoculture). Typically, plantation involves non-native or commercial species for timber, pulp, or other products. This work may be done in association of the Forest Department.

2.1.3.2.3 Horticulture

It includes to development of community horticulture and individual horticulture programmes specially on the barren and abandoned agricultural lands in the recharge areas by which the waste land of the watershed could be carpeted by plants to increase rainwater infiltration capacity of land. This work should be done in association with the local inhabitants and the Department of Horticulture and Watershed Management. This will also improve the economic base of the local inhabitants.

2.1.3.2.4 Development of Nurseries

It includes development of nurseries of broad leaved plants, fruits, herbs and grasses etc for conducting social forestry, community/ individual horticultural and herbal gardening programmes in the recharge zones. This work should be done in association of the local community and the Department of Horticulture and forest.

3.0 MICRO-PLANS FOR REJUVENATION MEASURES

Micro-plan maps (Fig.4, 5, 6 and 7) of the experimental area depict the notional locations/sites of mechanical and biological treatment measures. These maps may work as a faithful guide for implementation of micro-plans in field. A brief micro-plan related to proposed mechanical and biological treatments is presented in the following paragraphs.

3.1 Mechanical Treatment

Based on the standards of different mechanical treatment measures, geomorphological and geographical characteristics, numbers of each mechanical treatment measure were worked out which are presented in Table-3. A brief account of each mechanical treatment measure is described in the following paragraphs.

Table-1: Estimated numbers of mechanical treatment measures
table constructed in the experimental recharge area

Name of Mechanical Measures	Number of Treatment Mechanical Measures			
	Forest Land	Barren Land	Abandoned Agricultural Land	Agricultural Land
Infiltration Holes	1450000	1050000		-
Infiltration Trenches	4610	3468	-	-
Bio-plugs	-	-	20 across the upper reaches of two rills/gullies (see location of these rill/gullies in Fig.13)	
Rill/Gully –Plugs	45	-	5 across in the lower reaches of two rills/gullies (see location of these rill/gullies in Fig. 13)	
Terrace Ponds	-	2	10	20
Contour Bunds	-	-		5000 running meters
Check Dams	58	-	2 near the mouth of two rill/guiles (see location of these rill/gullies in Fig.12).	

3.1.1 Infiltration Holes

Infiltration holes are small holes (size 10-15 cm deep, 5cm Dia and density 5 per/m²) as shown in *Plate-VIII*. The purpose of these infiltration holes is to catch the excess hill slope runoff caused by high intensity rain drops and to facilitate it for groundwater recharge. These infiltration holes should be constructed in barren land, forest land and in abandoned agricultural land. Figure 4 depicts the notional locations of these proposed infiltration holes in the study area which must be constructed as much as 1450000 in forest land and 1050000 in barren land, and (Table-1). These holes may be filled after 2 monsoon rains, then new holes must be constructed. This work should be done in association with department of Forest and Watershed Management.

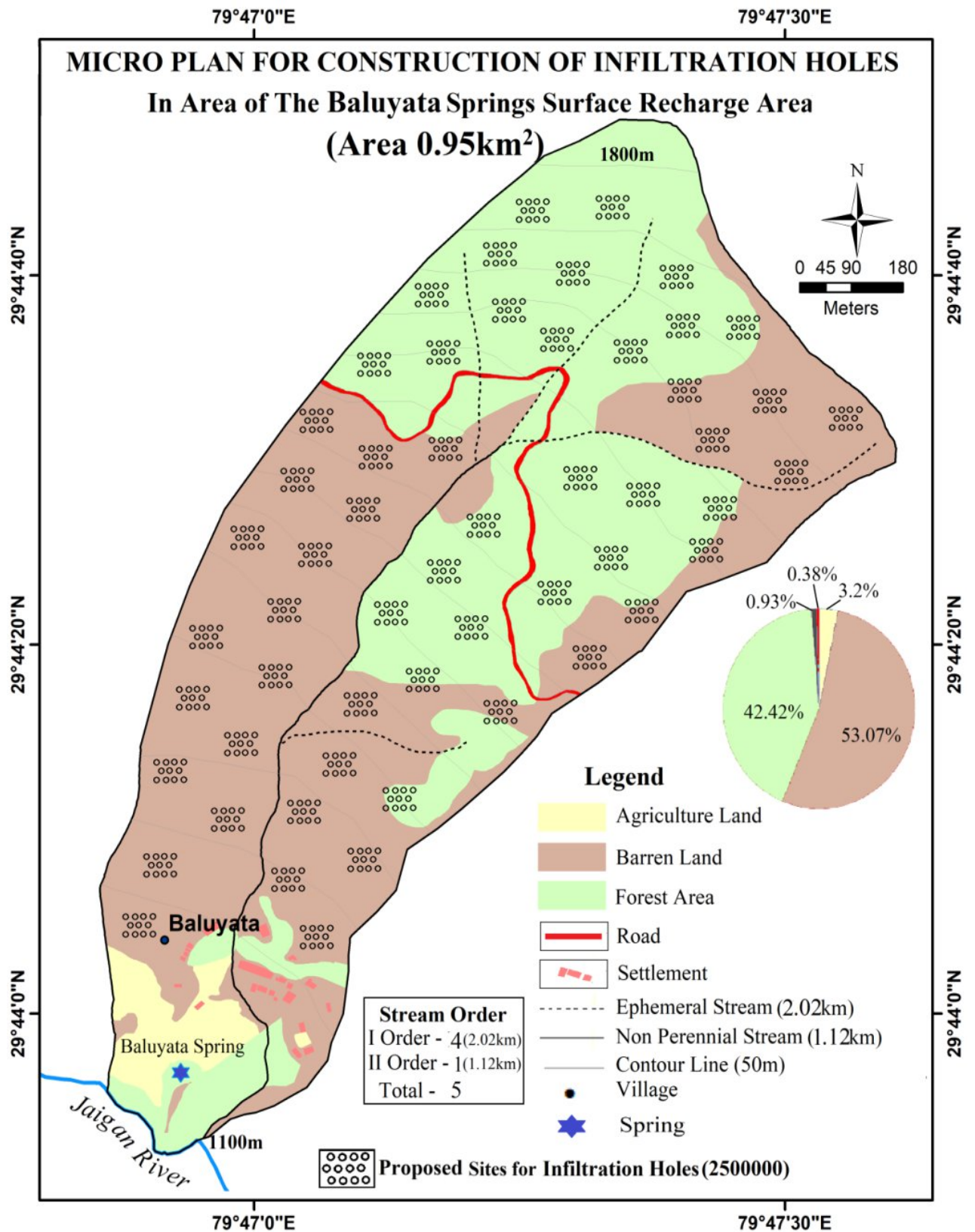


Fig.4: Micro-plan for construction of infiltration holes in the study area as per standards for Rain water harvesting to groundwater augmentation.

3.1.2 Infiltration Trenches

Infiltration trenches are small cuts (size 1.0 x 0.5 x 0.3 m) as shown in Plate-IX, on the hill slopes made at a spacing of 5m to 10m at an average slope condition of 20°. The distance of spacing may be decreased if the hill slope is low. These should be constructed in stagger patterns. The purpose of these infiltration trenches is to catch the excess hill slope runoff caused by high intensity rain drops and to facilitate it for groundwater recharge. These trenches must be constructed in the degraded forestland and barren lands preferably on convex hill slopes of barren land, degraded forest (Fig.5). A total of about 2341 trenches must be constructed, i.e., 4610 in forest land and 3468 in barren land (Table-1). These trenches may be filled after 3-4 monsoon rains. After filling it completely, a plant should be planted in this filled trench and new trench should be constructed in place of the filled trenches. This work should be done in association with department of Forest and Watershed Management.

3.1.3 Bio- Plugs

These are small bio-plugs to be constructed by vegetation leaves and dead wood logs or pine needles and soil across an ephemeral rill/gully. The purpose of these bio-plugs is to break the speed of runoff in the ephemeral rill/gully and to store rainwater to facilitate it for groundwater recharge (Plate-X). The size of these bio-plugs depends upon the local terrain condition. In the study area, there are two rills/gullies. About 45 bio-plugs at a vertical interval shall be needed across in the upper reaches of both the rills/gullies (Table-1, Fig.6). This work should be done in association with department of Forest and Watershed Management and Soil Conservation.

3.1.4 Rill Plugs/Gully Plugs

Due to high intensity rain events, the hill slopes are being eroded steadily through the runoff along ephemeral channels by which deep rills and gullies are being developed. To control this process of rilling/gullying, there is need to construct plugs of stones across these rills/gullies to decrease the velocity of streams, to store rainwater for infiltration and to trap the sediment. The size of such plugs (Plate-XI) depends upon the local terrain condition. In the study area 45 gully plugs shall be needed in the middle reaches across 4 ephemeral channels (Table-1, Fig.6). This work should be done in association with department of Watershed Management and Soil Conservation.

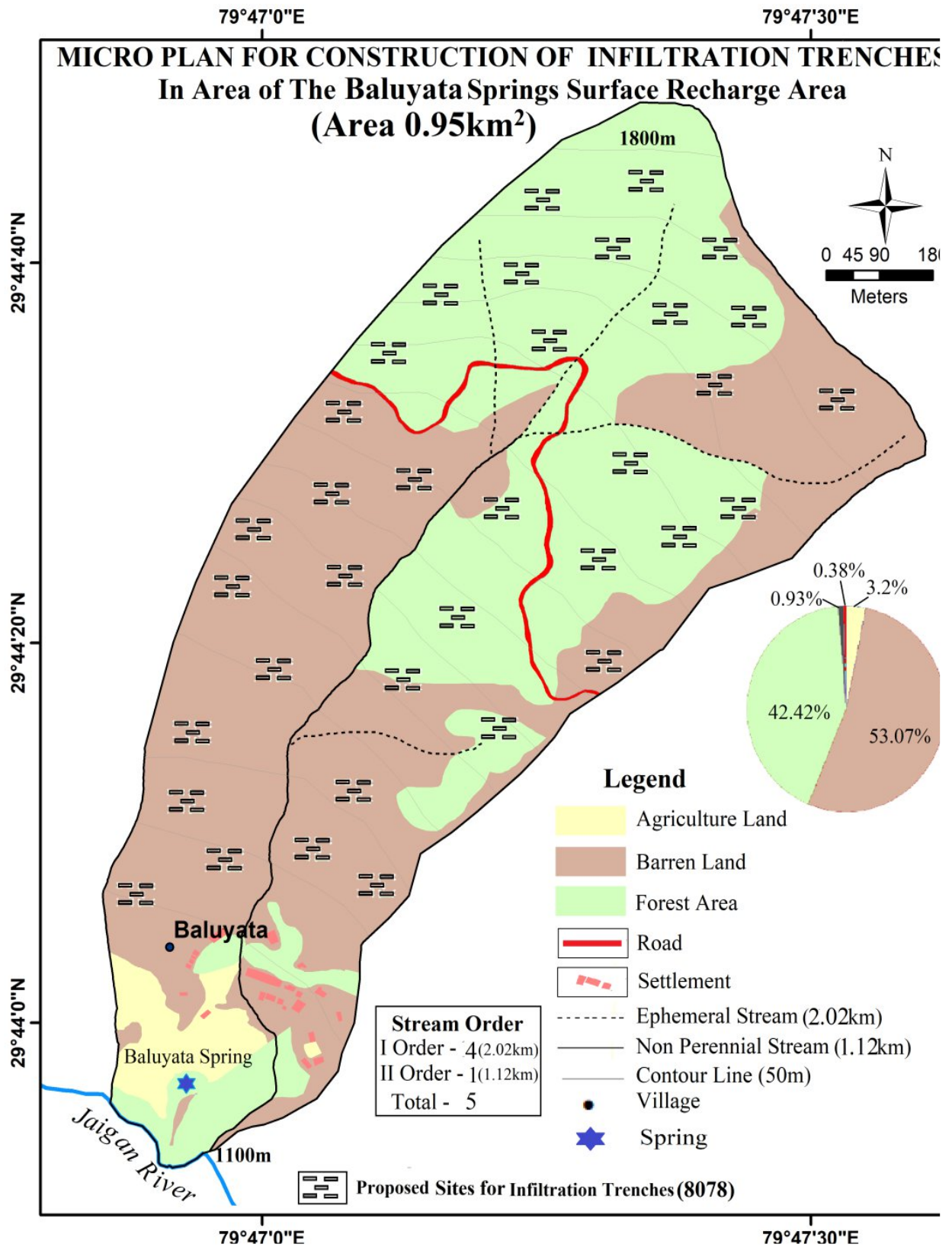


Fig.5: Micro-plan for construction of infiltration trenches in the study area as per standards for rainwater harvesting to groundwater augmentation.

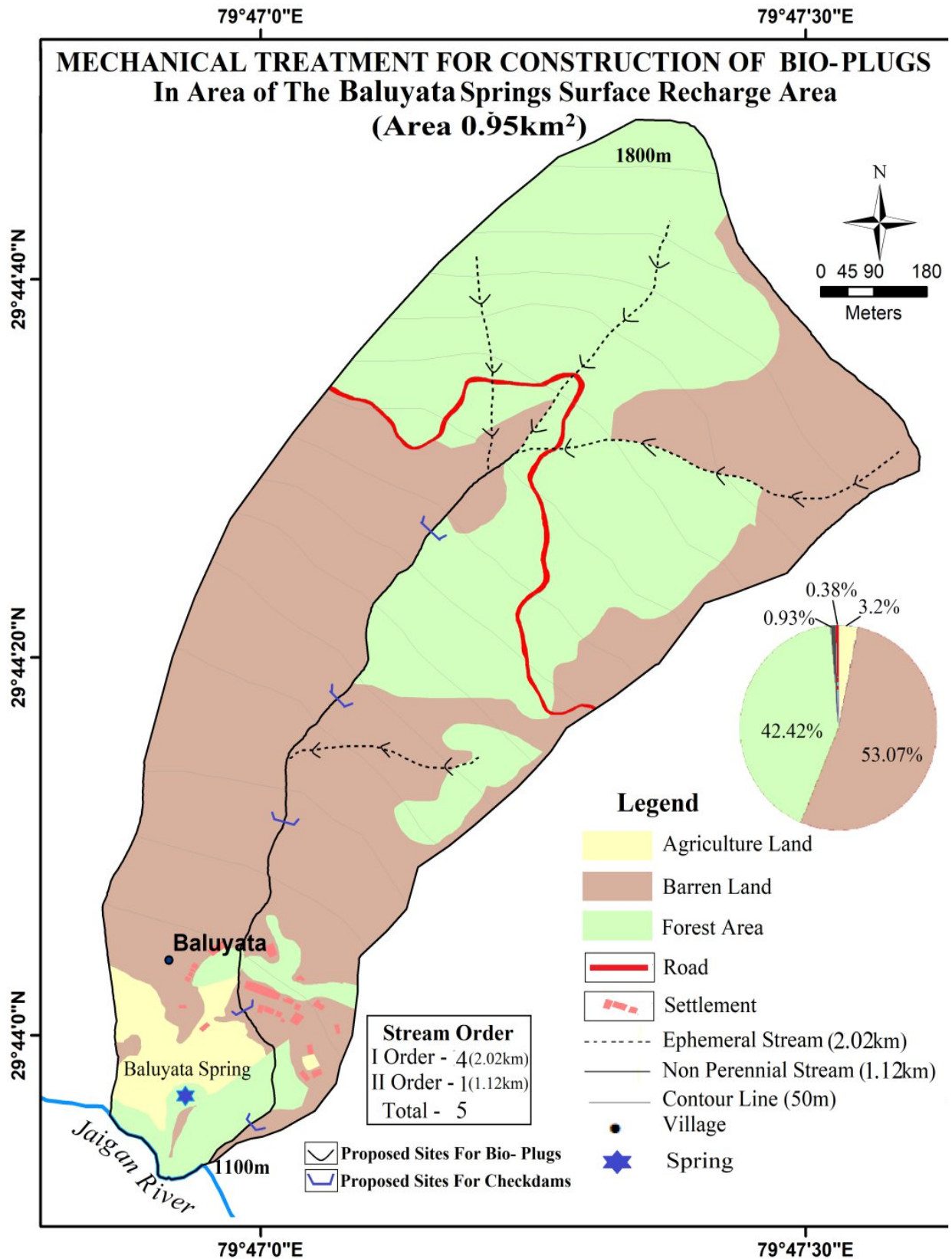


Fig.6: Micro-plan for construction of bio-plugs in the study area for rainwater harvesting to groundwater augmentation.

3.2 Biological Treatment

3.2.1 Afforestation and Plantation

Afforestation means establishing a forest in area where there was no previous tree cover. This involves planting of native species, allowing natural regeneration, and promoting biodiversity. This may be done on degraded land, like barren areas or abandoned agricultural lands. It is suggested that in the study area (Fig.7):

- i) about 0.50 km² land which constitutes about 53.11% of the study area is under barren condition. The spatial distribution of this barren land is presented in figure 6. This entire barren land may be used for i) afforestation of water harvesting (broad leaved) species, local shrubs and grasses such as Napier and vertebrae; and ii) plantation of a single species (monoculture) and non-native or commercial species for timber, pulp, or other products. This work may be done in association of the Forest Department.
- ii) About 0.50 km² of the study area which most of the part is as abandoned barren condition (Fig.2) which may be used for individual and community based horticultural farming in association with local community. This work should be done in association with the Department of Horticulture.
- iii) About 0.03km² land is under agricultural and barren condition in which on suitable sites nurseries of: i) broad leaf plants, local shrubs and grasses; ii) different herbal and ornamental plants; iii) different fruits; and flowers for floricultural development can be done in association with department of Forest and Horticulture.
- iv) should be done by initiating massive social forestry programmer in the recharge areas with close association with local community and the Forest Department and Watershed Management Department.

3.2.2 Development of Horticulture/Development of Floriculture

The experimental recharge area is well connected by road as the road passes from the center of the village, therefore there is good potentiality of development of horticulture as well as floriculture development in the experimental recharge area. The village panchayat should develop individual; and community horticultural plans and should identify land for implementing these plans, and should contact to the department for implementation. Similarly, the floriculture development plan should be done in association and guidance of the department of floriculture.

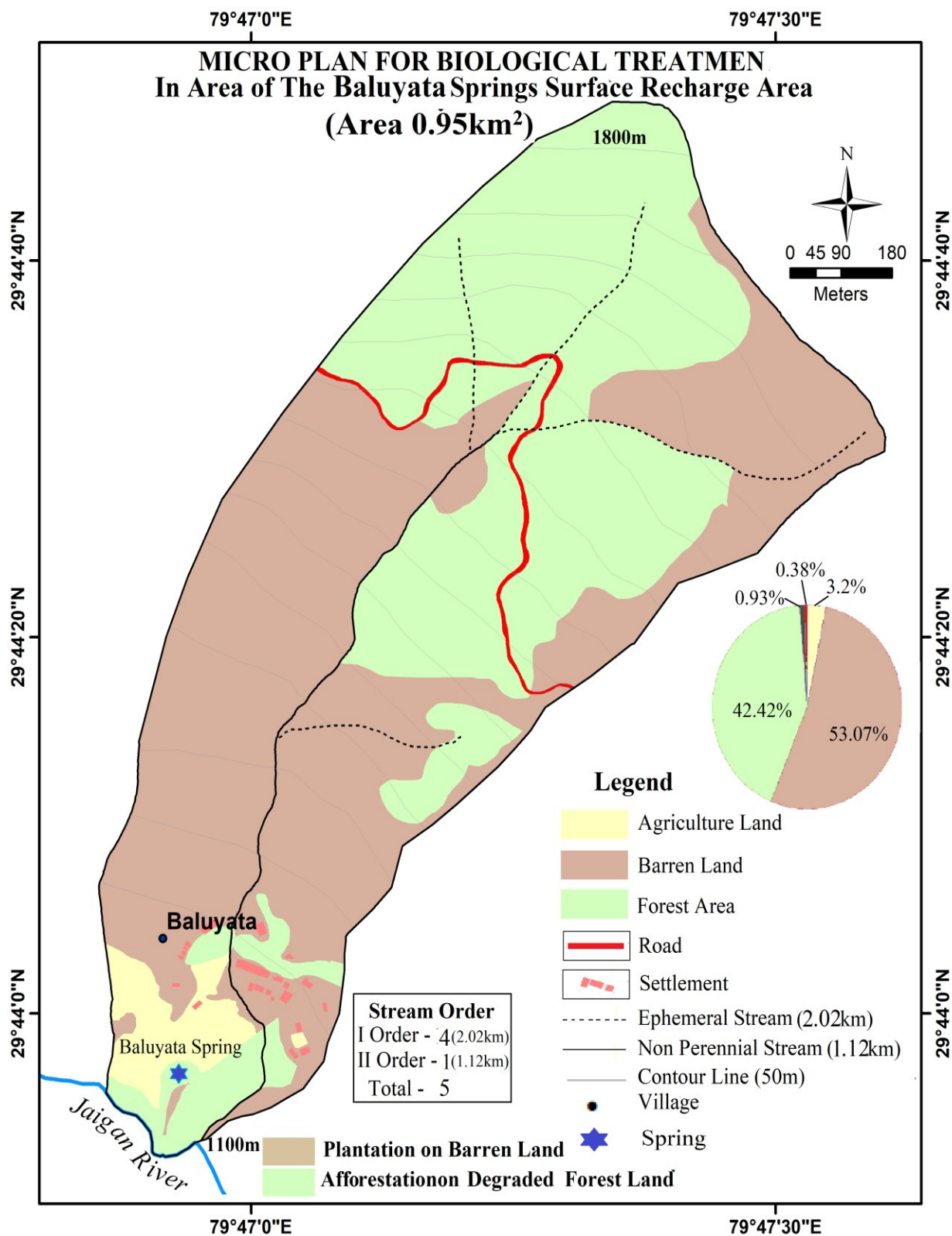


Fig.7: Micro-plan for biological treatment, i.e., afforestation and plantation in the study area for Development of Horticulture/Development of Floriculture.